

CS2610: Computer Organization and Architecture Lab

Lab 2: Nested Loops

9th February 2026

Objective

The goal of this lab is to get comfortable working with basic RISC-V concepts like nested loops, matrix memory layout.

Problem 1 : Bubble Sort

Write a program to sort an array of integers using the **Bubble Sort** algorithm. You must repeatedly compare adjacent elements and swap them if they are in the wrong order. Continue this process until the array becomes fully sorted.

- Store an integer array in memory.
- Sort the array using Bubble Sort logic.
- Display or dump the array contents before and after sorting.

Reference C Code

```
#include <stdio.h>

int main() {
    int arr[6] = {5, 2, 9, 1, 3, 6};
    int n = 6;

    for (int i = 0; i < n-1; i++) {
        for (int j = 0; j < n-i-1; j++) {
            if (arr[j] > arr[j+1]) {
                int temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
    return 0;
}
```

Problem 2: Matrix multiplication

Write a RISC-V assembly program to multiply two 3×3 matrices and store the result in a third 3×3 matrix.

Matrix Multiplication Rule

$$C[i][j] = \sum_{k=0}^2 A[i][k] \times B[k][j]$$

Hints

- Initialize all 9 elements in the result matrix to 0.
- Matrices are stored in row-major order.
- For an $n \times n$ matrix, the address of element $A[i][j]$ is:

$$\text{Address} = \text{Base} + (i \times n + j) \times 4$$

Example: Row-Major Memory Layout

Consider the following 3×3 matrix A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

In **row-major order**, elements are stored row by row in contiguous memory:

$$[1, 2, 3, 4, 5, 6, 7, 8, 9]$$

Assume:

- Base address of matrix $A = 0x1000$
- Each element = 4 bytes (word)

Worked Address Example

To compute the address of $A[2][1]$:

$$\text{Address} = \text{Base} + (i \times n + j) \times 4$$

$$= 0x1000 + (2 \times 3 + 1) \times 4 = 0x1000 + 28 = 0x101C$$

Resources

RISC-V Unprivileged Specification